

Measurement

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math). $\pi \approx 3.14$; π -values rounded to 1 d.p.

1. **[L1 · Rectangle area]** $A = l \times w = 9 \times 4 = 36 \text{ cm}^2$.
2. **[L1 · Square perimeter]** $P = 4 \times \text{side} = 4 \times 8 = 32 \text{ cm}$.
3. **[L1 · Triangle area]** $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$.
4. **[L1 · Circle circumference]** $C = 2\pi r = 2 \times 3.14 \times 7 = 44.0 \text{ cm}$.
5. **[L1 · Cube volume]** $V = \text{edge}^3 = 4^3 = 64 \text{ cm}^3$.
6. **[L1 · Cylinder volume]** $V = \pi r^2 h = 3.14 \times 2^2 \times 9 = 113.0 \text{ cm}^3$.
7. **[L2 · Parallelogram area]** $A = \text{base} \times \text{height} = 15 \times 6 = 90 \text{ cm}^2$.
8. **[L2 · Trapezoid area]** $A = \frac{1}{2}(a + b)h = \frac{1}{2}(10 + 6)(5) = 40 \text{ cm}^2$.
9. **[L2 · Composite area]** Rectangle $12 \times 5 = 60$; semicircle $r = 2.5$: $\frac{1}{2} \times 3.14 \times 2.5^2 = 9.8$.
Total $= 60 + 9.8 = 69.8 \text{ cm}^2$.
10. **[L2 · Prism surface area]** $SA = 2(lw + lh + wh) = 2(15 + 10 + 6) = 62 \text{ cm}^2$.
11. **[L2 · Cylinder surface area]** $SA = 2\pi r^2 + 2\pi rh = 2 \times 3.14 \times 4^2 + 2 \times 3.14 \times 4 \times 8$.
 $= 100.5 + 200.96 = 301.4 \text{ cm}^2$.
12. **[L2 · Cone volume]** $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}(3.14)(3^2)(7) = 65.9 \text{ cm}^3$.
13. **[L3 · Sphere volume]** $V = \frac{4}{3}\pi r^3 = \frac{4}{3}(3.14)(3^3) = 113.0 \text{ cm}^3$.
14. **[L3 · Sphere surface area]** $SA = 4\pi r^2 = 4 \times 3.14 \times 5^2 = 314.0 \text{ cm}^2$.
15. **[L3 · Pyramid volume]** $V = \frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \times 8^2 \times 9 = \frac{1}{3} \times 64 \times 9$
 $= 192 \text{ cm}^3$.
16. **[L3 · Volume with hole]** Outer: $3.14 \times 4^2 \times 10 = 502.4$; hole: $3.14 \times 2^2 \times 10 = 125.6$.
Remaining $= 502.4 - 125.6 = 376.8 \text{ cm}^3$.
17. **[L3 · Cone surface area]** $SA = \pi r^2 + \pi rl = 3.14 \times 5^2 + 3.14 \times 5 \times 13$.
 $= 78.5 + 204.1 = 282.6 \text{ cm}^2$.
18. **[L3 · Optimization]** For a fixed perimeter, area is greatest when the rectangle is a square.
Side $= 100 \div 4 = 25 \text{ m}$; area $= 25 \times 25 = 625 \text{ m}^2$.